SunMint Program — Consolidated Progress Report (v3)

Prepared by: Sophia Truesight (autopilot) · **For:** Gary Teh & TrueSight DAO members · **Date:** 2026-08-26

Revision note: v3 adds the **execution gap analysis** — everything still missing for the SunMint model to actually run (v2 added the mobile app layer).

1. The model has evolved — the mobile app is live

SunMint Farmer App (Android + iOS) — `TrueSightDAO/sunmint_mobile`, Capacitor 8, one codebase → native Android (signed APK) + iOS (TestFlight). 21+ merged PRs, UAT'd 2026-08-23.

Capability	Implementation
Tree planting reports	Live camera capture (JPEG q90, camera-only), species selector, GPS (8s, non-blocking)
Offline-first	Native SQLite queue + Filesystem photos; flush on reconnect
Identity & signing	RSA-2048, RSASSA-PKCS1-v1_5/SHA-256, byte-compatible web↔native; Keystore/Keychain
Ledger	Signed `[TREE PLANTING EVENT]` → Edgar → on-chain
Localization	PT default / EN toggle

Evolution: web app → native mobile → **next: tree-growth monitoring** — the phone becomes the farmer's carbon-MRV device.

2. Certification route (decided)

- **Pilot → Plan Vivo (PVCs)** — ACORN/CommuniTree standard; participatory monitoring; ≥60% revenue to communities
- **Scale → Verra VM0047 (ARR) + CCB** — ICVCM-approved; corporate offtake path
- VM0017 corrected (SALM ≠ ARR) — PR #294

3. Landmark precedent

Andean Cacao (Colombia) — first large-scale cacao agroforestry under Verra: VM0047 + CCB, Terra Global, **56,000+ VCUs first issuance**. SunMint is the same model.

4. Phone-based tree monitoring is proven

Program	What they do	Proof
ACORN (Rabobank)	Cocoa/coffee agroforestry, satellite + farmer apps, ~80% credit value to producers	Plan Vivo-certified
TREEO	Calibration-card DBH via phone camera	94–95% accuracy ($R^2 \geq 0.95$); Global Tree C-Sink Standard
Greenstand Treetracker	Geotagged repeat photos, ML-verified, paid per capture	500k+ trees, open-source
CommuniTree (Taking Root)	Farmer-verified survival & growth	28M trees, Plan Vivo since 2010

Method: calibration-card photo → DBH @1.3 m → allometric formulas → biomass → CO■e, satellite cross-checked.

5. MRV technology stack

Layer	Provider	Status
Farmer phone	SunMint Mobile App	■ live (monitor module = next)
Drone	PODream	■ discovery (no formal engagement)
Satellite	NOR Space	■ discovery (no quote commissioned)
Verification	VVB	■ not engaged

6. The decentralized model

Farmer-as-node · self-sovereign keys · on-chain attestation (DePIN+ReFi) · ACORN 80%-to-producers · Plan Vivo ≥60% · 10,000 ha can't be centrally surveyed → decentralization IS the scaling mechanism.

7. ■■ EXECUTION GAPS — what's still missing for the model to run

A. Certification & methodology gaps

#	Gap	Why it blocks	Owner	Unlock
A1	**PDD not written/validated**	No credits exist until validation passes	SunMint (we)	Develop PDD + engage VVB
A2	**No VVB engaged**	Validation/verification can't start	We hire	Shortlist + fee quote
A3	**Baseline land-use data absent**	VM0047 requires non-forest ≥10 yr proof	NOR Space	Commission potential analysis

#	Gap	Why it blocks	Owner	Unlock
A4	**Plot boundaries (GIS) not defined**	Terra RFP + PDD need boundaries	We + co-ops	Survey/boundary files

B. Data & MRV gaps

#	Gap	Why it blocks	Owner	Unlock
B1	**No per-tree identity in ledger**	Growth tracking needs individual records	DApp dev	Tree registry + IDs
B2	**monitor_tree_growth module not built**	Phone can't measure DBH/CO_2e yet	DApp dev	Build module (calibration-card + allometric)
B3	**No repeat measurements / history**	Growth curves = the credit evidence	DApp dev	Measurement history view
B4	**No field calibration plots**	Satellite/drone models need ground truth	PODream + co-ops	Establish plots
B5	**No satellite/drone data integration**	No cross-check layer live	NOR/PODream APIs	Integration hooks

C. Legal & tenure gaps

#	Gap	Why it blocks	Owner	Unlock
C1	**CAR land-tenure docs incomplete**	Terra RFP requires secured tenure	Co-ops	Collect CAR numbers
C2	**Land agreements not formalized**	Investors need land control proof	Legal	MOU/lease with co-ops
C3	**Brazil export-lane structure pending**	Revenue repatriation path	Legal (Elizabeth Wong/Onaya lane)	Entity setup

D. Community & farmer gaps

#	Gap	Why it blocks	Owner	Unlock
D1	**No formal community engagement record**	Plan Vivo + Terra require consent records	We + co-ops	Document meetings/FPIC
D2	**Farmer onboarding pipeline**	Need registered farmers per plot	Mobile app	Onboarding drive (app link + email)
D3	**Training on measurement method**	Data quality depends on farmer skill	We	Field training + TREEO-style guide

E. Financial gaps

#	Gap	Why it blocks	Owner	Unlock
E1	**Stage 0 capital (~$10–15k)**	NOR quote, legal, tenure work unfunded	DAO proposal	Capital injection (1 TDG/USD)
E2	**No offtake pre-signals**	Terra wants buyer interest	Business dev	Approach buyers post-PDD
E3	**NovaGaia pledge liability (–$510)**	Trees promised, unfinanced	Finance	Reconcile vs planted trees
E4	**No VVB fee budget**	Validation has a price tag	Finance	Budget in Stage 1

F. Governance & integration gaps

#	Gap	Why it blocks	Owner	Unlock
F1	**On-chain ↔ NOR integration undefined**	Decentralized edge must survive SaaS	Architecture	API/webhook contract
F2	**Data sovereignty terms absent**	Who owns biomass models/data?	Legal	Contract clause
F3	**PR #308 (community-MRV section) unmerged**	Whitepaper incomplete	Review	Merge + promote to prod

8. Financing plan (staged, source-mapped)

Stage	When	What it buys	Capital	Who pays
0 — De-risk & submit	0–3 mo	NOR feasibility, legal/tenure, Terra RFP	~$10–15k	DAO injection + grants (regen.fund, CocoaAction)
1 — PDD & Validation	3–12 mo	PDD, NOR baseline, VVB validation	~$45–80k	DAO round 2 + Terra RFP + chocolate co-funding
2 — Pilot Planting	12–36 mo	20–50 ha + maintenance + MRV ops	~$150–250k	Terra Climate Finance + offtake advances + PRONAF
3 — Scale to 10,000 ha	3–7 yr	Phased expansion (Bahia + Amazon)	$3–10M+	Forward VERR sales, Amazon Fund, Banco da Amazônia, Fundo Flora, impact capital

Principle: stage-gated — each stage's output de-risks the next source. No Stage 2 capex until PDD validated + offtake signed.

9. Terra RFP opportunity

- Open RFP for agroforestry (tropical, FCPF REDD+ — Brazil qualifies)
- **Min 3,000 ha** (phased OK) → frame at program scale, Bahia = Phase 1
- Readiness gaps = Section 7 items A3, A4, C1, D1, E1

10. Prioritized execution path (what to do next, in order)

1. **Merge + promote PR #308** (whitepaper complete) — 1 day
2. **Draft Stage 0 DAO capital proposal (~$10–15k)** — unlock everything below — 1 week
3. **Commission NOR Space potential analysis** (fills A3 + A4 + B5 at once) — 2–4 weeks
4. **Build monitor_tree_growth module in the mobile app** (B1–B3) — 2–3 weeks
5. **Shortlist VVBs** (A2) — 1 week, parallel
6. **Collect CAR tenure docs + community records** (C1, D1) via co-ops — 1–2 months
7. **Submit Terra RFP** once A3/A4/C1/D1/E1 met — target 2–3 months

Prepared by Sophia Truesight, autopilot · TrueSight DAO · Mission: restore 10,000 hectares of Amazon rainforest